Unit 7 Pre-Test

Quadratic Functions

Standards assessed: NC.M1.F-IF.4 · F-IF.7 · F-IF.9 · A-CED.1 · A-CED.2 · F-BF.1b

Name: _____ **Date:** _____ **Period:** _____

SCORE *3.0* **Basic Understanding**

NC.M1.F-IF.4 · NC.M1.F-IF.7 Read the key features of a parabola.

• *I can tell which way a parabola opens and find its vertex, axis of symmetry, y-intercept, and maximum or minimum.*

Answer each question. Use the picture to help you.

How to type your answers

point `(2, 5)` line `x=-1` expression `x^2+2x-2` inequality `y<=6`

1. *Direction*

Does $y = -2x^2 + 3$ open up or down?

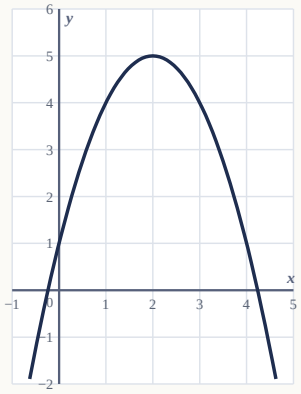
$-2x^2$ $+3$

$a = ?$

Answer: _____

2. *Vertex*

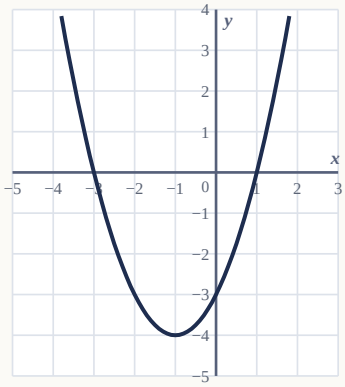
What is the vertex of the parabola?



Answer: _____

3. *Axis of symmetry*

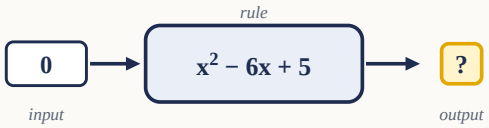
What is the axis of symmetry of the parabola?



x = _____

4. *y-intercept*

Find the y-intercept of $f(x) = x^2 - 6x + 5$.



Answer: _____

5. *Table of values*

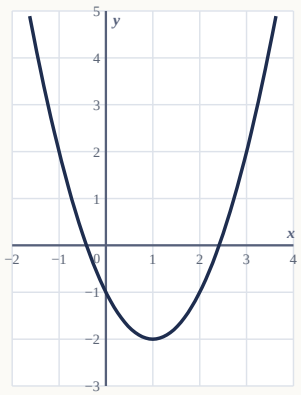
$f(x) = x^2 + 2x - 1$. Find f(3).

x	1	2	3
f(x)	2	7	?

f(3) = _____

6. *Max or min?*

Is the vertex of this parabola a maximum or a minimum?



Answer: _____

SCORE 4.0 Target Content

NC.M1.F-IF.4 · NC.M1.F-IF.7 · NC.M1.F-IF.9 · NC.M1.A-CED.2 · NC.M1.F-BF.1b Use forms of quadratics, compare functions, and combine functions.

• *I can find the vertex from standard or vertex form, give the range, compare two quadratics, write an equation, and add functions.*

Solve each problem. Show your work on paper.

7. *Find the vertex*

Find the vertex of $y = x^2 - 6x + 8$.

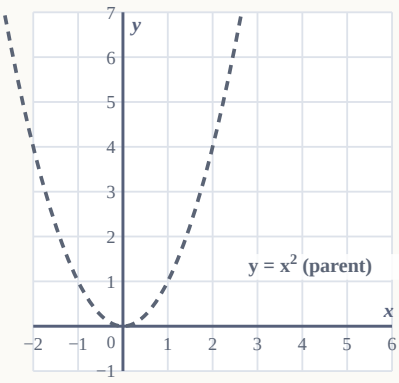
a = 1, b = −6, c = 8

$$x = \frac{-b}{2a} = ?$$

Answer: _____

8. *Vertex form*

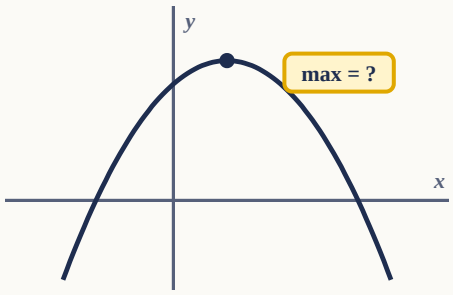
$y = (x - 4)^2 + 1$ is the parent graph $y = x^2$ moved. What is its vertex?



Answer: _____

9. *Range*

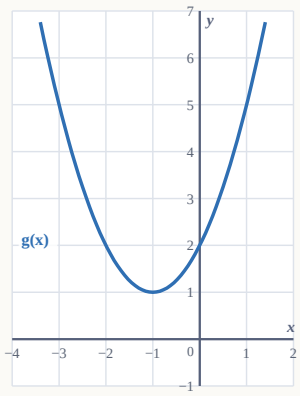
What is the range of $y = -(x - 1)^2 + 6$?



Answer: _____

10. *Compare*

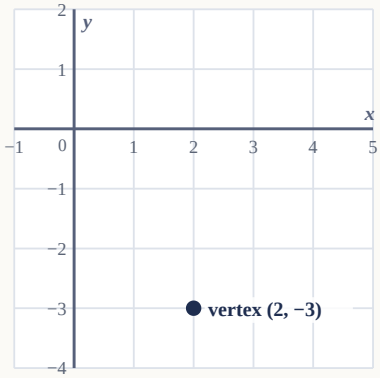
$f(x) = x^2 - 4x + 7$. The function g is graphed. Which function has the smaller minimum value: f or g?



Answer: _____

11. *Write the equation*

Write the equation of the parabola with vertex $(2, -3)$ and $a = 1$. Use vertex form.



$y =$ _____

12. *Add functions*

$f(x) = x^2 - 3$ and $g(x) = 2x + 1$. Write $(f + g)(x)$.

x	0	1	2
f(x)	−3	−2	1
g(x)	1	3	5
(f + g)(x)	?	?	?

$(f + g)(x) =$ _____

SCORE 5.0 Advanced Content

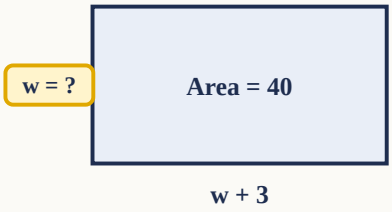
NC.M1.F-IF.4 · NC.M1.F-IF.9 · NC.M1.A-CED.1 · NC.M1.A-CED.2 · NC.M1.F-BF.1b Model with quadratics and build new functions.

• I can solve area and height problems, multiply functions, compare maximums, and write a quadratic from its zeros and a point.

Solve each problem. Some answers are words.

13. *Area equation*

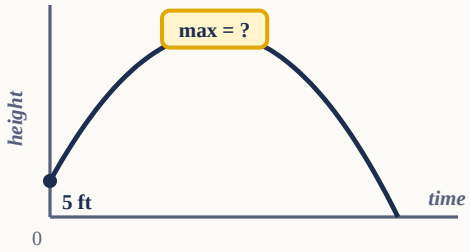
A rectangle's length is 3 more than its width. Its area is 40. Find the width.



w = _____

14. *Maximum*

$h(t) = -16t^2 + 64t + 5$ is a ball's height in feet. What is its **maximum** height?



feet = _____

15. *Multiply functions*

L(x) = x + 5 and W(x) = x − 2 are a rectangle's length and width. Write A(x) = L(x) · W(x) in standard form.

	x	+5
x	?	?
−2	?	?

A(x) = _____

16. *Compare*

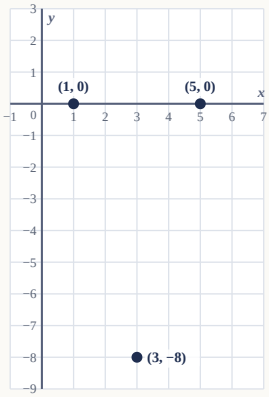
$f(x) = -x^2 + 4x$. The table shows g. Which has the greater maximum: f or g?

x	0	1	2	3	4
g(x)	1	4	5	4	1

Answer: _____

17. *Write the equation*

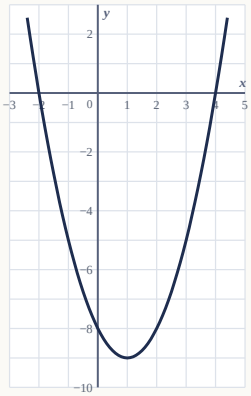
A parabola has zeros 1 and 5 and passes through (3, −8). Write its equation in standard form.



y = _____

18. *Decreasing*

$y = x^2 - 2x - 8$. For which x-values is the function **decreasing**?



Answer: _____